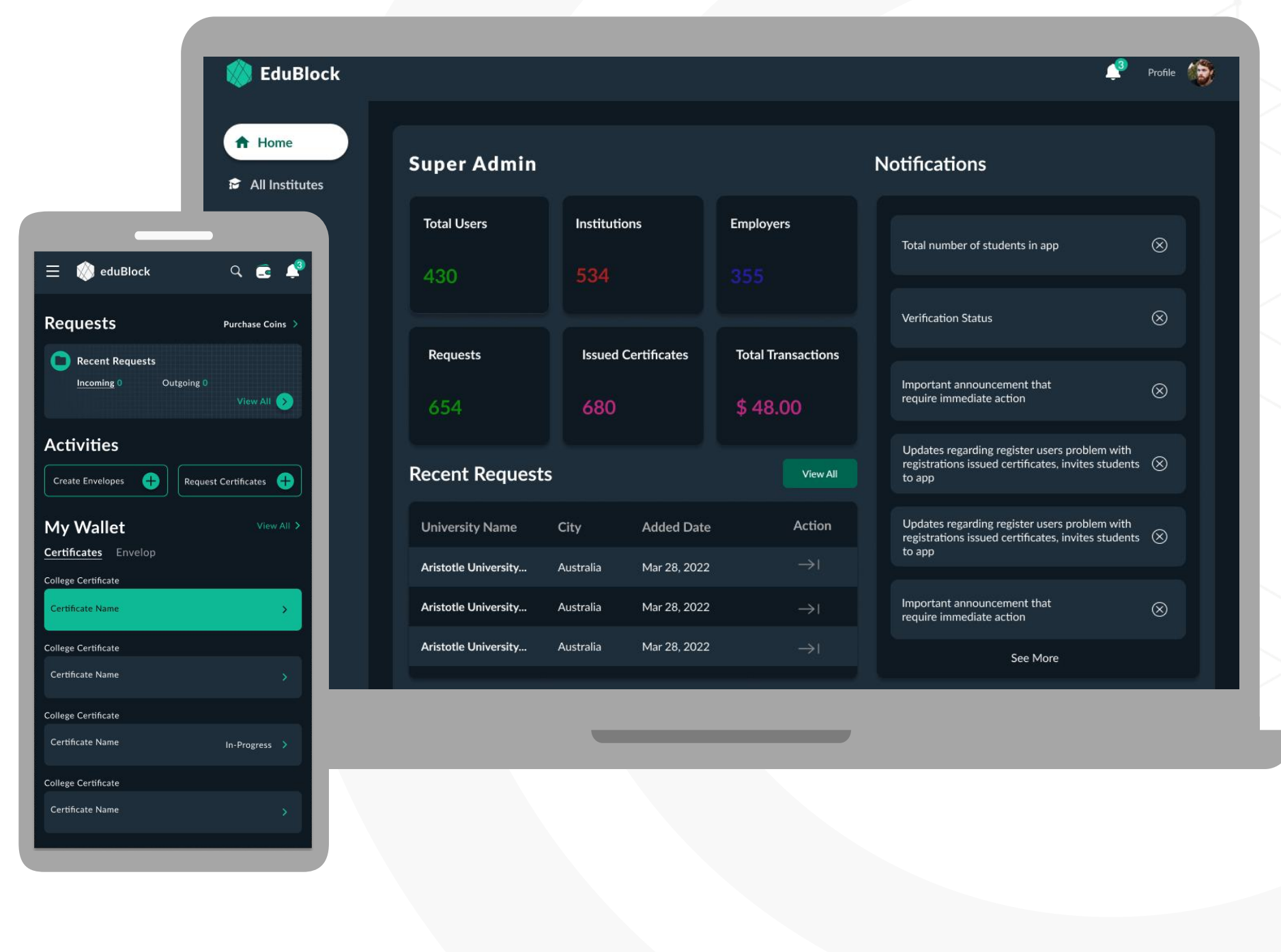


Case Study

EduBlock

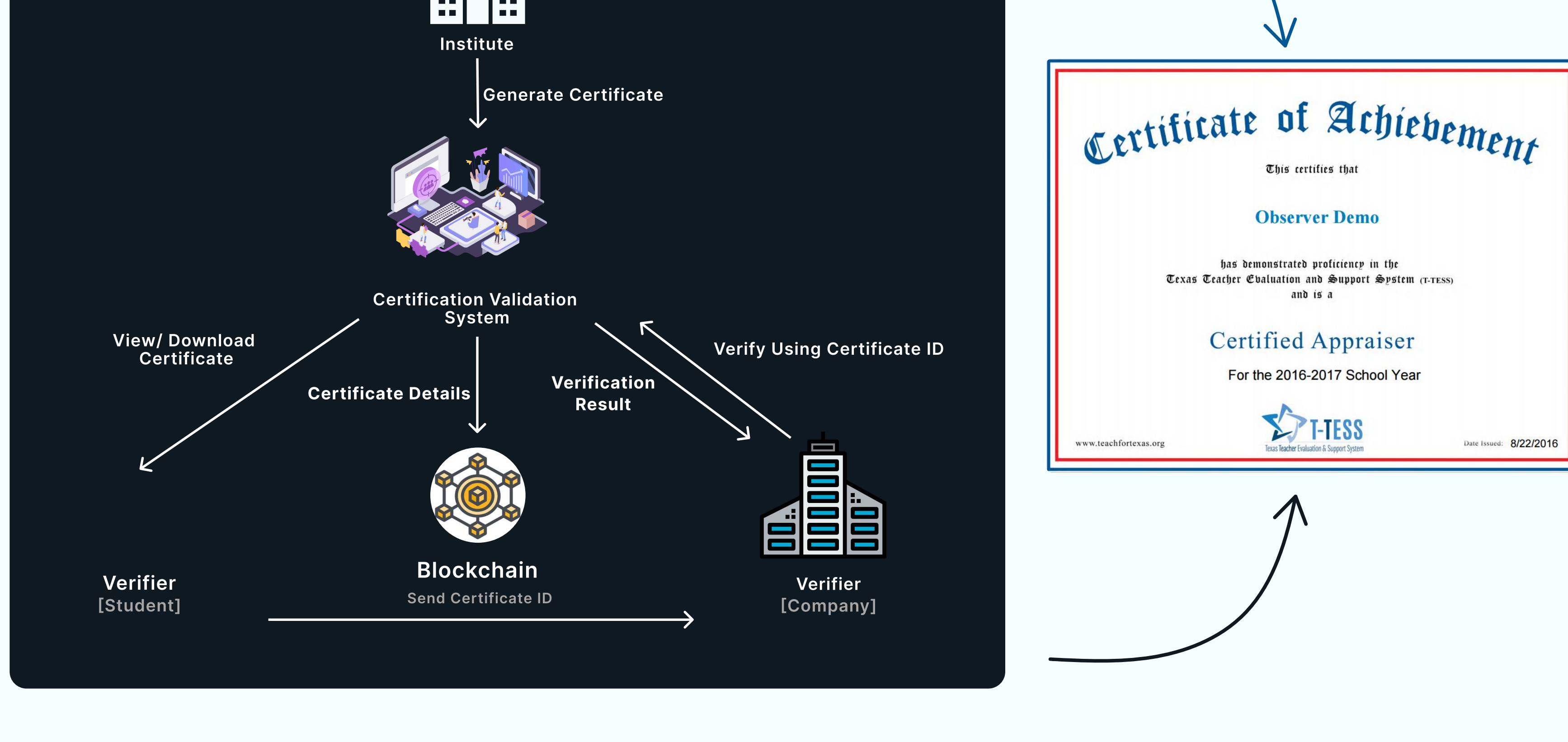
Secure certificate distribution via blockchain



Project Summary

EduBlock connects institutions, students, employers via blockchain. Enables certificate issuance, verification. Institutions manage departments, user roles, invitations, payment policies. Admins handle users, passwords. Graduates receive, employers verify certificates, ensuring network security.

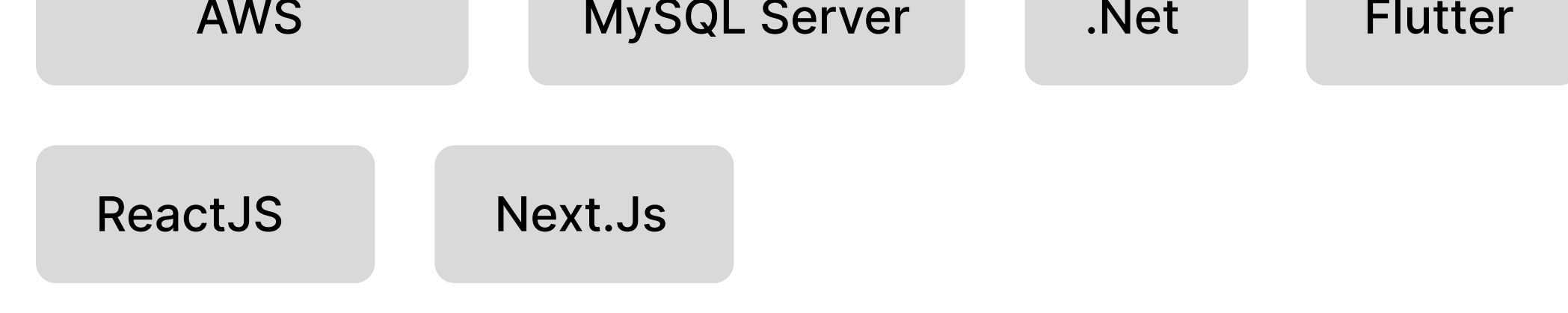
CONCEPT VISUALISATION



Services Used

- AWS / MySQL Server/.Net
- Micro Services Architecture And Development
- CI/CD Implementation And Management

Technology Stacks



Objective

The primary goal of this system is to provide a robust, secure, and user-friendly platform to manage digital identities and verifiable credentials for academic achievements. By leveraging Decentralized Identifiers (DID), the system intends to bridge the gap between universities, students, and corporate entities, thereby streamlining the verification of academic qualifications.

Key Features

- Decentralized Identifiers (DID)**
The core of the system is built around DIDs. Each actor (university, company, student) will possess a unique DID, enabling smooth identity management and ensuring data integrity.
- DID issuer (University) Dashboard**
Tailored for institutions, this dashboard facilitates the creation of DIDs for students, designing certificate templates, and issuing verifiable credentials. Enhanced with roles & permissions, it provides granular control, ensuring only authorized personnel can execute specific actions.
- Company Dashboard**
Exclusively for companies, this dashboard focuses on API key management and subscription tracking. The integrated system allows companies to easily verify the credentials presented to them by prospective candidates or partners.
- Super Admin Dashboard**
A permissioned portal to oversee and manage the entire system. It provides functionalities like creating new university or company profiles, monitoring DIDs, and observing the credentials issued across the system.
- API & Documentation**
Extensive APIs allow seamless integration with external systems, ensuring scalability and wider application. Detailed documentation ensures stakeholders can quickly understand and leverage the system's capabilities.
- Credential Holder (Student) Mobile App**
A convenient tool for students to view their DIDs, manage their credentials, and securely share them through various methods, including email, QR codes, and deep links.
- Enhanced Security**
Recognizing the sensitive nature of data, the platform incorporates two-factor authentication (2FA) across all dashboards. Additionally, login records ensure traceability, providing an added layer of security.

eduBlock Benefits

- Streamlined Verification Process**
Companies can instantly verify the academic credentials of potential employees, reducing the hiring cycle and minimizing fraudulent claims.
- Empowering Students**
With the mobile app, students can have instant access to their credentials, share them on-demand, and maintain control over their data.
- Enhanced Data Integrity**
By using DIDs and blockchain-backed verifiable credentials, the system ensures that the data is tamper-proof and trustworthy.
- Cost-effective**
The platform reduces administrative burdens on universities, cuts verification costs for companies, and minimizes credential fraud, resulting in significant long-term savings.

Challenges

Manual verification is time-consuming and costly, and paper certificates are prone to fraud. Without a centralized database, verification is resource-intensive. Implementing blockchain, issuing cryptographic digital certificates, integrating university systems with third-party platforms, using smart contracts, developing verification apps, collaborating with governments, and educating stakeholders can streamline and secure the verification process.

Eliminating Manual Certification Processes

1. Targeting time and cost savings.
2. Utilizing blockchain technology for efficiency gains.
3. Transitioning from manual to automated certificate issuance.

Streamlining Employer-Student Communication

1. Create user-friendly system for employers to request certificates.
2. Facilitate smooth and secure communication between employers and students.
3. Implement subscription plan for employers or third parties.
4. Buy coins through subscription to verify certificates.
5. Utilize blockchain technology for certificate verification process.

Ensuring Certificate Security and Uniqueness

1. Implementing a system with unique, blockchain-generated certificate identifiers.
2. Ensuring authenticity and proof of ownership through DID.
3. Utilizing blockchain technology to safeguard against fraud.
4. Each certificate assigned a non-mutable identifier.
5. Providing security and authenticity assurance through DID.

Enabling Instant Certificate Verification

1. Unique DID assigned to each certificate ensures authenticity.
2. Blockchain technology generates non-mutable identifiers for certificates.
3. DID serves as proof of ownership, safeguarding against fraud.
4. Each certificate linked to a secure, immutable identifier.
5. System ensures authenticity and integrity of issued certificates.

Our Solutions

Blockchain solutions play a pivotal role in addressing the multifaceted challenges of certificate issuance and verification. By leveraging blockchain technology, institutions can ensure the security, authenticity, and integrity of certificates within a vast ecosystem of educational stakeholders, including administrators, students, employers, and regulatory bodies.



SOLUTIONS

Eliminating Manual Certification Processes

1. Managing Institution Admins, Department Admins, and Institute Users.
2. centralized databases, blockchain, digital signatures. Effective revocation management ensures trust.
3. Maintaining trust necessitates effective revocation management against fraud or misconduct.
4. Role management and data synchronization present notable challenges in certificate systems.

Flexible Payment Policy Integration

1. Institutions may enforce varied payment approaches, from free to cost-per-transaction models.
2. Developing a payment system that integrates smoothly with certificate processes is essential.
3. Managing diverse payment models poses scalability challenges.
4. Balancing payment flexibility with user-friendly interfaces is crucial.

Ensuring Security and Authenticity in Certificate Systems.

1. Secure Private Keys: Protecting private keys is vital to prevent forgery.
2. Trust in Issuers: Establishing trust in issuing institutions is crucial.
3. Reliable Verification: Implement robust verification methods for authenticity.
4. Effective Revocation: Manage revocations effectively to maintain trust.

Adoption and Interoperability

1. Achieving agreement among educational institutions, employers, and students is crucial.
2. Ensuring compatibility with existing systems is vital.
3. Standardizing protocols facilitates seamless integration.
4. Efficient operation relies on seamless integration and standardized protocols.

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New Age Tech Builders

As a pioneering tech provider, we harness the most advanced technology stacks like Flutter, Go, Rust, Solidity and more.

Security-First Approach

Our commitment to security is ingrained in our "secure-by-design" philosophy.

Culture of Innovation

Innovation is our cornerstone. Our team embraces a culture of continuous learning and experimentation

Innovative and Tech-Savvy

Confident and Charismatic

Solution Oriented

Entrepreneurial Spirit

Thankyou